ETALE SHEAVES SATISFY DESCENT ALONG SURJECTIVE INTEGRAL, PROPER, OR FPPF-FLAT MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 816-828 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 830-863, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_spaces-pushouts.json (pb=1450, stmt 816-828, proof 830-863).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GFT: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 729–752. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GFV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 792–801. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09YC: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 863–863. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04WP: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 863–863. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MH: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 863–863. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04ZH: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 863–863. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03HC: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 863–863. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which has one of the following properties: surjective and integral, surjective and proper, or surjective and flat and locally of finite presentation. Then the functor*

$$Sh(Y_{étale}) \longrightarrow \text{descent data for étale sheaves wrt } \{X \to Y\}$$

is an equivalence of categories.

Proof. Observe that the base change of a proper surjective morphism is proper and surjective, see Morphisms of Spaces, Lemmas 04WP and 03MH. Hence by Lemma 0GFT we may work étale locally on Y. Hence we reduce to Y being an affine scheme; some details omitted.

Assume Y is affine. By Lemma 0GFV it suffices to find a morphism $X' \to X$ where X' is a scheme such that $X' \to Y$ is surjective and integral, surjective and proper, or surjective and flat and locally of finite presentation.

In case $X \to Y$ is integral and surjective, we can take $X = X'$ as an integral morphism is representable.

If f is proper and surjective, then the algebraic space X is quasi-compact and separated, see Morphisms of Spaces, Section 03HC and Lemma 04ZH. Choose a scheme X' and a surjective finite morphism $X' \to X$, see Limits of Spaces, Proposition 09YC. Then $X' \to Y$ is surjective and proper.

Finally, if $X \to Y$ is surjective and flat and locally of finite presentation then we can take an affine étale covering $\{U_i \to X\}$ and set X' equal to the disjoint $\coprod U_i$. □